

CS420 - Stock Portfolio Management System

Relational Schema

Schema Overview

The following five relations make up the database. Primary keys are marked (PK) and foreign keys are marked (FK).

INVESTOR(InvestorID PK, FirstName, LastName, Email, Country, Phone)
BROKERAGE_ACCOUNT(AccountID PK, AccountType, BrokerName, Balance, InvestorID FK)
COMPANY(CompanyID PK, CompanyName, Industry, Country, FoundedYear)
STOCK(StockID PK, TickerSymbol, ExchangeName, CurrentPrice, CompanyID FK)
TRADE_TRANSACTION(TransactionID PK, TradeDate, TradeType, Quantity, PricePerShare, AccountID FK, StockID FK)

Table Definitions

INVESTOR

Stores basic information about each investor in the system.

Column	Data Type	Constraints	Description
InvestorID	INT	NOT NULL, AUTO_INCREMENT, PK	Unique investor identifier
FirstName	VARCHAR(50)	NOT NULL	First name
LastName	VARCHAR(50)	NOT NULL	Last name
Email	VARCHAR(100)	NOT NULL	Email address
Country	VARCHAR(50)	NOT NULL	Country of residence
Phone	VARCHAR(20)	NULL	Phone number (optional)

Foreign Key(s): None

BROKERAGE_ACCOUNT

Represents a financial account used by an investor for trading.

Column	Data Type	Constraints	Description
AccountID	INT	NOT NULL, AUTO_INCREMENT, PK	Unique account identifier
AccountType	VARCHAR(50)	NOT NULL	e.g. Individual, Roth IRA, 401k

Column	Data Type	Constraints	Description
BrokerName	VARCHAR(100)	NOT NULL	Name of the brokerage firm
Balance	DECIMAL(15,2)	NOT NULL, DEFAULT 0.00	Current account balance
InvestorID	INT	NOT NULL, FK	References INVESTOR(InvestorID)

Foreign Key(s): InvestorID -> INVESTOR(InvestorID)

COMPANY

Stores information about companies whose stocks can be traded.

Column	Data Type	Constraints	Description
CompanyID	INT	NOT NULL, AUTO_INCREMENT, PK	Unique company identifier
CompanyName	VARCHAR(100)	NOT NULL	Full name of the company
Industry	VARCHAR(100)	NOT NULL	Business sector or industry
Country	VARCHAR(50)	NOT NULL	Country where company is based
FoundedYear	INT	NULL	Year the company was founded (optional)

Foreign Key(s): None

STOCK

Represents a stock issued by a company.

Column	Data Type	Constraints	Description
StockID	INT	NOT NULL, AUTO_INCREMENT, PK	Unique stock identifier
TickerSymbol	VARCHAR(10)	NOT NULL	Stock ticker symbol e.g. AAPL
ExchangeName	VARCHAR(50)	NOT NULL	Exchange where stock is listed
CurrentPrice	DECIMAL(10,2)	NOT NULL	Current market price
CompanyID	INT	NOT NULL, FK	References COMPANY(CompanyID)

Foreign Key(s): CompanyID -> COMPANY(CompanyID)

TRADE_TRANSACTION

Records every buy or sell trade made through a brokerage account.

Column	Data Type	Constraints	Description
TransactionID	INT	NOT NULL, AUTO_INCREMENT, PK	Unique transaction identifier
TradeDate	DATE	NOT NULL	Date the trade was made
TradeType	VARCHAR(4)	NOT NULL, CHECK (BUY or SELL)	Either BUY or SELL
Quantity	INT	NOT NULL	Number of shares traded
PricePerShare	DECIMAL(10,2)	NOT NULL	Price per share at time of trade
AccountID	INT	NOT NULL, FK	References BROKERAGE_ACCOUNT
StockID	INT	NOT NULL, FK	References STOCK(StockID)

Foreign Key(s): AccountID -> BROKERAGE_ACCOUNT(AccountID), StockID -> STOCK(StockID)

Foreign Key Summary

Table	Foreign Key	References	Meaning
BROKERAGE_ACCOUNT	InvestorID	INVESTOR(InvestorID)	Each account belongs to one investor
STOCK	CompanyID	COMPANY(CompanyID)	Each stock is issued by one company
TRADE_TRANSACTION	AccountID	BROKERAGE_ACCOUNT(AccountID)	Each trade is made through one account
TRADE_TRANSACTION	StockID	STOCK(StockID)	Each trade involves one stock